

Promoting Ethical, Sustainable, and Trustworthy Practices in Digital Media Platform Design

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Abstract—Digital media platforms increasingly rely on interface patterns that capture and exploit users’ attention, often at the expense of well-being, autonomy, and trust. Previous research has mapped fifteen Attention-Capture Damaging Patterns (ACDPs), highlighting their socio-technical implications and potential harms. Building on this foundation, this study moves beyond diagnosing harmful patterns to explore constructive, innovative alternatives. We propose and analyze a set of ethical design directions aimed at mitigating the negative effects of ACDPs while promoting sustainable and trustworthy interaction practices. Our analysis is structured around four guiding dimensions—augmented autonomy and agency, cognitive sustainability, authenticity and trust, and community and belonging—which together provide a rationale for rethinking design choices in alignment with human-centered values. By articulating how innovative alternative design patterns can reinforce these dimensions, we contribute to the broader agenda of ethical digital interaction and offer reflections to support both academic debate and practical adoption by technology companies.

Keywords—attention-capture damaging patterns, digital media platforms, ethical UI design, cognitive sustainability, user agency, community and belonging, digital well-being

I. INTRODUCTION

Digital platforms, particularly social media, have become central to how people communicate and consume content [15]. These platforms offer unprecedented opportunities for personal expression, social interaction, knowledge sharing, and social curation [14, 17]. However, alongside these benefits, there has been a rise in design strategies that prioritize engagement over digital well-being [9-12]. Known as Attention-Capture Damaging Patterns (ACDPs), these design practices exploit cognitive and emotional vulnerabilities to maximize time spent on platforms, often at the cost of user autonomy, cognitive sustainability, trust, and meaningful social interaction.

The concept of ACDPs was initially introduced by Roffarello and colleagues [1], who identified eleven patterns that exploit users’ cognitive and emotional vulnerabilities and proposed alternative design strategies to counteract their harmful effects. These design mechanisms are rooted in a

platform-centric model that prioritizes user retention and engagement over individual well-being [4, 5]. They are engineered to exploit cognitive and psychological vulnerabilities, subtly steering user behavior in ways that often lead to diminished autonomy, heightened anxiety, and social disconnection [6, 7].

Building upon existing research in intelligent user interfaces, Gomes et al. [2] mapped a complementary set of fifteen ACDPs. Examples of the mapped ACDPs include “Like and Share Pressure”, “Infinite Scroll”, and “Autoplay of Shared Reels”, all of which encourage compulsive engagement and reduce the user’s ability to make deliberate choices. While a few items overlap with the earlier typology, their contribution was to broaden the repertoire of harmful practices, propose additional alternative patterns, and organize the fifteen ACDPs into a taxonomy composed of three categories: *Cognitive Manipulation and Interface Ambiguity Patterns*, *Compulsive Engagement Patterns*, and *Disruption of Healthy Use Patterns*. This taxonomic framing was also addressed in [8] and enriches our analytical toolkit for understanding how attention manipulation operates across digital media platforms and highlights targeted opportunities for socio-technical mitigation.

These studies collectively demonstrate that while ACDPs can enhance short-term engagement metrics, they often undermine long-term digital well-being, contributing to cognitive fatigue, social anxiety, and superficial interactions. Despite growing recognition of these harms, current design guidelines for mitigating ACDPs remain disparate and fragmented. While recent work has proposed frameworks focused on proactive attention support [3] and manipulative design practices [16], there is a lack of a holistic, actionable framework that integrates ethical, cognitive, social, and creative considerations for digital media platform designers.

Motivated by these gaps, the present work seeks to provide an analysis of ACDPs and their alternative patterns, structured around four core dimensions. By adopting this framework, we aim to guide designers toward digital experiences that balance engagement with ethical, deliberate, and socially responsible practices.

Specifically, this paper makes the following contributions:

- **Taxonomy of ACDPs and alternative patterns.** We review fifteen distinct ACDPs and their corresponding alternative patterns designed to mitigate these negative effects.
- **Dimension-based analysis.** We connect each pattern and alternative to the four core dimensions, illustrating how design choices impact user autonomy, cognition, trust, and social belonging.
- **Design rationale for ethical interaction.** We articulate a framework that enables designers to create engaging digital systems that support digital well-being, promote meaningful interactions, and ensure long-term sustainability.

This work provides actionable guidance for designing digital environments that respect users’ attention, cognitive resources, and social connections, offering a path toward more ethical and healthy interaction design.

The rest of the paper is described as follows. Section II elaborates on the four core dimensions for ethical and healthy interaction design that form the conceptual framework for our study. Section III presents a detailed discussion of the alternative patterns for ethical and healthy interactions, demonstrating how each is specifically engineered to counteract the identified damaging patterns and address the core dimensions. Section IV summarizes the main findings, discusses the current limitations, and proposes pathways for future research in the field of ethical user interface (UI) design.

II. CORE DIMENSIONS FOR ETHICAL AND HEALTHY INTERACTION DESIGN

Designing digital platforms that respect users’ attention, autonomy, and digital well-being requires a clear framework to guide ethical interaction. Drawing on a prior analysis of ACDPs and their alternatives (see [2] for more details), we identified four core dimensions that serve as a foundation for designing healthier and more responsible digital experiences: *Augmented Autonomy and Agency*; *Cognitive Sustainability*; *Authenticity and Trust*; and *Community and Belonging*. Each dimension addresses a distinct aspect of user experience while collectively promoting ethical, sustainable, and empowering interactions.

Our analysis of ACDPs revealed that coercive design primarily manifests by eroding users’ free will, suppressing the authentic presentation of self and diminishing psychological autonomy. Since ACDPs fundamentally work by hijacking the user’s *intentionality*, suppressing authentic *expression* and eroding their *independence*, prioritizing these three concepts acts as a critical conceptual lens, guaranteeing the ethical integrity and alignment of the four core dimensions against these manipulative practices. This aligns with recent literature proposing an ontological view of digital well-being, suggesting that establishing a healthier connection with technology necessitates intentional choices, such as the strategic decision to ‘switching off to switch on’, to fundamentally improve one’s relationship with digital tools [18]. The selection of intentionality, expression, and independence as fundamental conceptual characteristics is rooted in the essential human capacities most directly targeted and degraded by ACDPs.

The relationship between these characteristics and the dimensions of our framework is intrinsically reinforcing. Intentionality is the operational principle of the dimension “Augmented Autonomy and Agency”. Digital autonomy is only truly restored when a user’s actions result from conscious choice rather than impulsive, engineered triggers. Similarly, independence is essential not just for autonomy but also for ensuring “Cognitive Sustainability”, as it protects the user from having their attention and mental resources hijacked or coerced into unintended actions. Furthermore, genuine expression serves as the foundation for both “Authenticity and Trust” and “Community and Belonging”. Trust is established when the platform facilitates the presentation of an authentic self, and a healthy sense of belonging thrives in environments where individuals can express themselves freely, without the fear of social coercion or manipulative feedback loops. These three characteristics define the necessary preconditions for the core dimensions to succeed in practice, as depicted in Fig. 1.

Dimension Characteristic	Augmented Autonomy and Agency	Cognitive Sustainability	Authenticity and Trust	Community and Belonging
 Intentionality	Users control engagement	Mindful decisions, not compulsive	Transparency of information source	Meaningful, not shallow, interactions
 Expression	Opportunities for content creation	Focus and flow supported	Honest feedback and signals	Inclusion and diversity fostered
 Independence	Meaningful choices, no manipulation	Managing cognitive load effectively	Reliable and consistent interactions	Shared purpose and mutual support

Fig. 1. Core dimensions for ethical and healthy interaction design.

A. Augmented Autonomy and Agency

This dimension refers to the design principle of enhancing users’ capacity to make deliberate choices, express themselves creatively, and control their digital interactions. It goes beyond simple “creative freedom” to empower users with tools, feedback, and interfaces that augment their decision-making, personal goals, and engagement with content. Key aspects include:

- **Intentional control.** Users decide when, how, and with what intensity to engage with digital content.
- **Creative expression.** Platforms provide opportunities to produce, remix, or reflect on content rather than passively consuming it.
- **Enhanced autonomy.** Systems support meaningful choices without imposing manipulative attention-capture mechanisms.
- **Sustained engagement with purpose.** Users can pursue goals aligned with their personal values or creative interests, instead of being driven by compulsive patterns.

These aspects are critical because many ACDPs reduce user agency by encouraging compulsive or reactive behaviors. Alternative design patterns such as “Reflective Interactions” or “Opt-In Viral Challenges” are intended to restore autonomy and allow users to engage in ways that are deliberate, meaningful, and aligned with their goals.

B. Cognitive Sustainability

Cognitive sustainability focuses on preserving and supporting users' mental resources during digital interactions. It emphasizes creating interfaces and experiences that avoid cognitive overload, attention fragmentation, and mental fatigue while enabling users to engage in a sustainable, healthy, and deliberate manner. Here we highlight some particularly noteworthy aspects:

- **Managing cognitive load.** Interfaces prevent excessive information at once, reducing stress and enhancing comprehension.
- **Supporting focus and flow.** Platforms facilitate uninterrupted engagement and allow users to concentrate on meaningful tasks.
- **Promoting mindful engagement.** Interaction design encourages conscious decisions rather than compulsive or automatic behavior.
- **Sustaining long-term mental energy.** Digital tools should help users maintain clarity, reflection, and creativity over time, avoiding burnout or cognitive exhaustion.

Alternative patterns such as “Neutral Headlines” and “Batch Notifications” are designed to reduce mental strain and protect attention, promoting deliberate and conscious engagement [13]. In contrast, some original patterns may induce overstimulation or compulsive interactions.

C. Authenticity and Trust

This dimension ensures that digital platforms provide reliable, transparent, and genuine interactions, fostering users' confidence in the platform and its content. It emphasizes reducing deceptive practices, misinformation, and manipulative design elements that can erode trust and compromise user judgment. Key elements include:

- **Transparency.** Clear indication of the source and nature of information, notifications, or recommendations.
- **Honest feedback and signals.** Metrics, rewards, and alerts accurately reflect actual user activity or platform states.
- **Reliable interactions.** Users can anticipate outcomes of their actions and trust that the platform behaves consistently.
- **Protection from manipulation.** Design prevents exploitative or deceptive patterns that could undermine user autonomy, decision-making, or social relationships.

All of the alternative patterns we propose contribute directly to authenticity and trust, ensuring that users can engage confidently with digital platforms. For example, “Accurate and Relevant Notifications” and “Personal Progress Track” reinforce predictability and truthfulness, contrasting with manipulative patterns such as deceptive notifications or emotionally manipulative headlines.

D. Community and Belonging

Community and belonging emphasize fostering meaningful social connections, collaboration, and a sense of inclusion within digital environments. It supports positive interpersonal interactions, social cohesion, and user participation while avoiding coercive or manipulative engagement. Essential elements include:

- **Meaningful social interactions.** Genuine communication and collaboration rather than shallow engagement.
- **Inclusion and diversity.** Users feel represented and respected within communities.
- **Shared purpose and mutual support.** Platforms enable collaboration, group achievements, and constructive feedback.
- **Positive social reinforcement.** Engagement cues support encouragement instead of pressure or manipulation.

TABLE I. THE 15 MAPPED ACDPs [2].

Original Pattern (ACDP)	Alternative Pattern
P1. Like and Share Pressure	Reflective Interactions
P2. Emotionally Manipulative Headlines	Neutral Headlines
P3. FOMO Alerts	COMOs Prompts
P4. Friend Suggestion Disguised as Invitation Notification	Clear and Distinct Notifications
P5. Deceptive Notifications in Archived Groups	Accurate and Relevant Notifications
P6. Streak Rewards	Flexible Rewards
P7. Infinite Comments Scroll	Comment Limit
P8. Gamified Engagement Metrics	Personal Progress Track
P9. Highlight-Reels on Social Media	User-Controlled Playback
P10. Autoplay of Shared Reels	Press-to-Play
P11. Automatic Story Transitions	User-Triggered Transitions
P12. Pushed Viral Challenges	Opt-In Viral Challenges
P13. Endless Exploration Tabs	Topic-Specific Exploration
P14. Endless Notifications and Engagement-Based Notifications	Batch Notifications
P15. Time-Wasting Challenges	Purposeful Challenges

Although generally harmful to autonomy, creativity, or cognitive sustainability, some original patterns can still foster a sense of belonging or social interaction, albeit in a manipulative or superficial way. For instance, patterns such as “Like and Share Pressure” or “Streak Rewards” may create social bonds or perceived community through repetitive interactions, but they do so at the cost of user agency, making their effect controversial. In contrast, our alternative patterns are designed

to foster genuine, positive interactions, promoting belonging ethically, consciously, and aligned with user well-being.

Together, these four dimensions—Augmented Autonomy and Agency, Cognitive Sustainability, Authenticity and Trust, and Community and Belonging—provide a holistic framework for ethical and healthy interface design. They address the user’s control and creative freedom (Agency), the preservation of cognitive resources (Sustainability), the trustworthiness of the platform (Authenticity), and the quality and meaningfulness of social interactions (Community). By explicitly considering these dimensions, designers can mitigate ACDPs while promoting deliberate, responsible, and beneficial engagement in digital environments. Table I presents all 15 original patterns along with their corresponding alternative design strategies.

III. ALTERNATIVE PATTERNS FOR ETHICAL AND HEALTHY INTERACTIONS

Building on the analysis of the fifteen mapped ACDPs, we proposed a set of alternative patterns focused on promoting digital well-being and integrity. Each of these patterns aims to mitigate the negative effects of ACDPs by directly addressing the four critical dimensions of our framework. Below, we detail how each alternative pattern operates within these dimensions.

Reflective Interactions. In opposition to the pressure of “Like and Share Pressure” (P1), the “Reflective Interactions” pattern encourages a more deliberate interaction style. It enhances *Augmented Autonomy and Agency* by empowering the user to make a conscious decision to comment or engage in a discussion, rather than simply reacting with a like or a share. From a *Cognitive Sustainability* standpoint, this approach reduces the overload of superficial engagement notifications, allowing the user to focus on deeper, more meaningful interactions with less stress. This pattern also strengthens *Authenticity and Trust* because interactions are based on genuine content and thoughtful reflection, not on inflated metrics. As a result, a sense of *Community and Belonging* becomes more authentic and profound, as connections are built on real discussions.

Neutral Headlines. Countering the manipulative nature of “Emotionally Manipulative Headlines” (P2), the “Neutral Headlines” pattern prioritizes objectivity. This approach returns *Augmented Autonomy and Agency* to the user, allowing them to decide if the content is relevant based on a clear and honest title, rather than a manipulated emotional reaction. *Cognitive Sustainability* is directly improved as the pattern alleviates mental strain and emotional fatigue by eliminating constant exposure to psychological triggers. For the *Authenticity and Trust* dimension, transparency in content presentation is fundamental, as the user learns to trust that the platform will not deceive them. This also fosters *Community and Belonging* as discussions become grounded in facts and information rather than emotional impulses, leading to a more informed and cohesive community.

COMO Prompts. In contrast to the alerts that generate “FOMO Alerts” (P3), the “Comfortable Missing Out (COMO) Prompts” pattern acts as a tool to promote disconnection. It reinforces *Augmented Autonomy and Agency* by empowering the user to make the conscious decision to disconnect without

feeling guilt or pressure. In terms of *Cognitive Sustainability*, this pattern directly alleviates the anxiety and stress associated with concerns about missing information online, allowing the user to redirect their attention to the real world. For the *Authenticity and Trust* dimension, the platform demonstrates that it genuinely cares about the user’s digital well-being, not just treating them as a metric to be maximized. *Community and Belonging* are also positively impacted, as a culture of healthy disconnection is valued, promoting a healthier sense of belonging.

Clear and Distinct Notifications. Countering the ambiguous notifications of “Friend Suggestion Disguised as Invitation Notification” (P4), the “Clear and Distinct Notifications” pattern promotes transparency. It enhances *Augmented Autonomy and Agency* by giving users full control over which notifications to open, without being misled by a deceptive design. *Cognitive Sustainability* is improved as the pattern reduces the mental burden of deciphering the true meaning of each notification, allowing the user to direct their attention more efficiently. *Authenticity and Trust* are strengthened because the platform becomes transparent, and the user learns that the platform will not deceive them to induce clicks. For *Community and Belonging*, the pattern improves the quality of social engagement since users can distinguish between genuine interactions and automated alerts.

Accurate and Relevant Notifications. The “Accurate and Relevant Notifications” pattern directly counters the misdirection inherent in “Deceptive Notifications in Archived Groups” (P5), ensuring that platform communication remains truthful and user-centric. This design choice powerfully restores *Augmented Autonomy and Agency* by fostering user trust. That is, individuals can rely on each alert to signify a genuine interaction, not a deliberate tactic to lure them back into the platform. Regarding *Cognitive Sustainability*, the pattern significantly preserves the user’s mental resources and attention span by eliminating the needless stress and interruption caused by false alerts. Furthermore, *Authenticity and Trust* are substantially strengthened because the platform is committed to providing truthful, verifiable information about user activity. Consequently, *Community and Belonging* are enhanced through higher-quality social communication, as accurate notifications ensure that interactions are meaningful and relevant to the user.

Flexible Rewards. In opposition to the rewards system of “Streak Rewards” (P6), the “Flexible Rewards” pattern promotes a healthier, less coercive usage. It enhances *Augmented Autonomy and Agency* by allowing users to define their own usage goals, rather than being forced into continuous daily use to earn rewards. *Cognitive Sustainability* is directly benefited, as the pattern reduces the anxiety and pressure of maintaining a streak, contributing to a more conscious and less mentally exhausting use. This pattern also increases *Authenticity and Trust*, as the reward system demonstrates that it cares about the user’s health and not about manipulation to increase screen time metrics. For *Community and Belonging*, removing social competition over engagement metrics allows the community to focus on healthy usage goals and mutual support.

Comment Limit. Instead of the continuous flow of “Infinite Comments Scroll” (P7), the “Comment Limit” pattern offers a more controlled flow of content. It returns *Augmented Autonomy and Agency* to the user, allowing them to consciously decide when and how much to consume, rather than being swept up in the endless flow. *Cognitive Sustainability* is mitigated because the pattern provides natural stopping points in the comment stream, allowing the user to rest and reflect. The pattern also strengthens *Authenticity and Trust* by demonstrating that the platform respects the users’ ability to disconnect and does not force them to continue consuming content indefinitely. This promotes *Community and Belonging* by encouraging more focused and high-quality discussions.

Personal Progress Track. In contrast to engagement metrics that gamify interaction from “Gamified Engagement Metrics” (P8), the “Personal Progress Track” pattern redirects the focus to individual development. It empowers the user, reinforcing their *Augmented Autonomy and Agency* by allowing them to focus on self-determined personal goals, removing the pressure of comparing themselves with the engagement of others. *Cognitive Sustainability* is directly benefited, as the pattern alleviates the mental strain and anxiety generated by social competition. *Authenticity and Trust* are also enhanced because the user receives an honest and personalized metric of progress, without the distortion of social comparison. The pattern fosters a more supportive and less competitive *Community and Belonging*, where the focus is on personal growth.

User-Controlled Playback. Instead of the endless video stream from “Highlight-Reels on Social Media” (P9), the “User-Controlled Playback” pattern gives the user control over video consumption. This returns *Augmented Autonomy and Agency* to the user, allowing them to decide whether or not to play a video. Regarding *Cognitive Sustainability*, the pattern reduces stress and unintentional consumption, as the user can prepare for the content or simply ignore it without interruption. *Authenticity and Trust* are reinforced because the platform demonstrates that it respects the user’s choice to consume the content or not. The pattern also promotes *Community and Belonging* by allowing users to share videos more intentionally, leading to more meaningful conversations.

Press-to-Play. The “Press-to-Play” pattern is a variation of the previous one, countering the automatic playback of shared videos from “Autoplay of Shared Reels” (P10). It grants full control over content consumption, requiring a conscious action (a click) to start playback. *Cognitive Sustainability* is protected as the pattern prevents unintentional consumption and mental fatigue caused by automatic video playback. *Authenticity and Trust* are strengthened because the platform prioritizes user choice over automatic engagement. For *Community and Belonging*, the pattern fosters more authentic social interactions, as users only share what they want to be seen.

User-Triggered Transitions. Instead of automatic transitions between stories from “Automatic Story Transitions” (P11), the “User-Triggered Transitions” pattern returns control to the user. It reinforces *Augmented Autonomy and Agency* by allowing users to decide when and where story navigation goes. In terms of *Cognitive Sustainability*, the pattern provides

natural stopping points in content consumption, allowing the user to stop and reflect. This pattern also increases *Authenticity and Trust* by ensuring the navigation system is honest and user-controlled. In line with this, it promotes more meaningful connections and fosters *Community and Belonging* by allowing the user to focus on one story at a time, which leads to deeper interactions with friends.

Opt-In Viral Challenges. In opposition to the forceful nature of “Pushed Viral Challenges” (P12), the “Opt-In Viral Challenges” pattern promotes voluntary participation. It ensures that the user participates in viral challenges by deliberate choice, not by social pressure or algorithmic manipulation. When considering *Cognitive Sustainability*, the pattern reduces the anxiety and social pressure of participating in every viral trend, allowing users to maintain focus and intention. *Authenticity and Trust* are reinforced as the platform values user consent over viral engagement metrics. The pattern also fosters a healthier sense of *Community and Belonging*, where participation is a choice.

Topic-Specific Exploration. Countering the infinite discovery tabs from “Endless Exploration Tabs” (P13), the “Topic-Specific Exploration” pattern offers a clear focus. It grants users complete control over their content discovery journey, allowing them to decide what they want to explore and for how long. In terms of *Cognitive Sustainability*, the pattern alleviates information overload and mental fatigue, as content is limited and relevant to the chosen topic. This pattern increases *Authenticity and Trust* by ensuring the platform provides content transparently. It also promotes a sense of *Community and Belonging* by allowing users to explore specific topics of shared interest.

Batch Notifications. In opposition to constant notifications from “Endless Notifications and Engagement-Based Notifications” (P14), the “Batch Notifications” pattern promotes grouping non-urgent alerts. It returns *Augmented Autonomy and Agency* to the user, allowing them to decide when and how to interact with notifications. When considering *Cognitive Sustainability*, the pattern directly contributes to reducing stress and attention overload, allowing the user to focus on their daily tasks without constant interruptions. *Authenticity and Trust* are strengthened because the platform demonstrates that it respects the user’s time and does not monitor them to induce interruptions. The pattern also reduces the social pressure to respond immediately, fostering more meaningful and intentional interactions with implications for *Community and Belonging*.

Purposeful Challenges. In contrast to the time-consuming nature of “Time-Wasting Challenges” (P15), the “Purposeful Challenges” pattern offers activities with clear benefits. It empowers users, reinforcing their *Augmented Autonomy and Agency* by engaging in activities with clear purpose and goals. *Cognitive Sustainability* is achieved because engagement is directed toward productive activities that provide measurable benefits in mitigating mental fatigue and cognitive drain. *Authenticity and Trust* are reinforced as the platform demonstrates that it cares about helping users achieve goals. This fosters a supportive *Community and Belonging* to shared objectives.

IV. CONCLUSION, LIMITATIONS, AND FUTURE WORK

The pervasive nature of ACDPs has degraded the quality of digital interaction, leading to diminished user autonomy and agency and a strain on cognitive sustainability. This paper addressed this critical challenge by proposing a structured, socio-technical framework aimed at shifting design paradigms from extractive engagement to ethical well-being. Our core contribution lies in defining four essential dimensions for ethical and healthy interaction design, which provide a comprehensive lens for evaluating and mitigating harmful design practices. We then instantiated this framework by analyzing fifteen alternative patterns for ethical and healthy interactions, with each pattern systematically mapped to counteract a detrimental ACDP. This work establishes a foundational guide for researchers and practitioners committed to developing human-centric digital media platforms.

While this study offers a robust conceptual framework and a practical catalog of alternative patterns, it is subject to certain limitations that must be acknowledged. For instance, the present work is primarily conceptual and theoretical. Although the mappings are grounded in established psychological and behavioral science theories, their efficacy has yet to be proven through large-scale empirical testing. Moreover, the catalog of ACDPs addressed here is limited to 15 prominent patterns, and the digital landscape continuously evolves. The biggest hurdle for implementation is the challenge of convincing big tech companies to adopt the alternative patterns. Their current business models are deeply entrenched in and financially dependent upon the very extractive engagement that ACDPs generate, creating a fundamental tension between digital well-being and corporate profit.

These limitations, however, naturally set the stage for several crucial avenues of future research and development. The most critical next step involves empirical validation, where we intend to conduct controlled user studies to quantitatively measure the impact of the alternative patterns on key user metrics such as perceived autonomy and reported stress levels. This data will be essential to provide the regulatory and economic evidence needed to counter the profit-driven resistance of major platform providers. Finally, the research agenda includes expanding the catalog of patterns, requiring a systematic effort to monitor emerging digital trends and develop corresponding alternative patterns to ensure the framework remains comprehensive and relevant in the long term.

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